

Solutions to Computer Network Assignment 3

I. Go-Back-N (GBN) Protocol

(1) Possible Sequence Number Set in the Sender's Window

The receiver is currently expecting the k -th packet, which means all packets with sequence numbers less than k have been successfully acknowledged. The GBN protocol uses a fixed-size continuous sender window with a size of 4.

Thus, the starting sequence number of the sender's window (base) can range from k to $k+3$ (if the base exceeded $k+3$, the window would go beyond the scope of unacknowledged packets, violating the protocol logic). The possible sequence number sets are: $[k, k+1, k+2, k+3]$, $[k+1, k+2, k+3, k+4]$, $[k+2, k+3, k+4, k+5]$, and $[k+3, k+4, k+5, k+6]$, with all sequence numbers falling within the range 1~1024.

(2) Possible Values of the ACK Field

The ACK field indicates the sequence number of the next packet the receiver expects to receive. GBN uses in-order acknowledgment, so the receiver will not send out-of-order ACKs.

The minimum ACK value is k , which occurs when the receiver has not yet received the k -th packet (only acknowledging packets before k). The maximum ACK value is $k+4$, which happens when the receiver has successfully received all four packets ($k, k+1, k+2, k+3$) in the sender's window and now expects $k+4$.

Between these two extremes, the receiver may have received 1 to 3 in-order packets, corresponding to ACK values of $k+1, k+2$, and $k+3$. Therefore, the possible values of the ACK field are $k, k+1, k+2, k+3$, and $k+4$.

II. SDN OpenFlow Flow Table Configuration

(1) Flow Table Entries for Switch S1

Matching Conditions	Action
Input Port=4, Source IP=10.2.0.3 (h3), Destination IP=10.3.0.5 (h5)	Output to Port 1
Input Port=4, Source IP=10.2.0.3 (h3), Destination IP=10.3.0.6 (h6)	Output to Port 1
Input Port=4, Source IP=10.2.0.4 (h4), Destination IP=10.3.0.5 (h5)	Output to Port 1
Input Port=4, Source IP=10.2.0.4 (h4), Destination IP=10.3.0.6 (h6)	Output to Port 1
Input Port=4, Source IP=10.3.0.5 (h5), Destination IP=10.2.0.3 (h3)	Output to Port 4
Input Port=4, Source IP=10.3.0.5 (h5), Destination IP=10.2.0.4 (h4)	Output to Port 4
Input Port=4, Source IP=10.3.0.6 (h6), Destination IP=10.2.0.3 (h3)	Output to Port 4
Input Port=4, Source IP=10.3.0.6 (h6), Destination IP=10.2.0.4 (h4)	Output to Port 4
Input Port=1 or 4, Destination IP=10.1.0.1 (h1)	Output to h1's access port
Input Port=1 or 4, Destination IP=10.1.0.2 (h2)	Output to h2's access port
Source IP=10.1.0.1 (h1), Destination IP=10.1.0.2 (h2)	Output to h2's access port
Source IP=10.1.0.2 (h2), Destination IP=10.1.0.1 (h1)	Output to h1's access port

(2) Flow Table Entries for Switch S3 (Firewall Behavior)

Matching Conditions	Action
Source IP=10.1.0.2 (h2), Destination IP=10.3.0.5 (h5)	Allow Forwarding (Output to the corresponding port)
Source IP=10.1.0.2 (h2), Destination IP=10.3.0.6 (h6)	Allow Forwarding (Output to the corresponding port)
Source IP=10.2.0.3 (h3), Destination IP=10.3.0.5 (h5)	Allow Forwarding (Output to the corresponding port)
Source IP=10.2.0.3 (h3), Destination IP=10.3.0.6 (h6)	Allow Forwarding (Output to the corresponding port)

Matching Conditions	Action
Source IP=10.1.0.1 (h1), Destination IP=10.3.0.5 (h5)	Drop
Source IP=10.1.0.1 (h1), Destination IP=10.3.0.6 (h6)	Drop
Source IP=10.2.0.4 (h4), Destination IP=10.3.0.5 (h5)	Drop
Source IP=10.2.0.4 (h4), Destination IP=10.3.0.6 (h6)	Drop

III. Packet Scheduling Algorithms

(1) Priority Scheduling (Class 1 > Class 2 > Class 3)

Scheduling Rule: Prioritize Class 1 packets, followed by Class 2, and then Class 3. For packets of the same class, transmit them in the order of arrival. One packet is transmitted per time slot.

Time Slot t	Transmitted Packet	Reason
1	1	Class 2, arrival time 0.35 (the earliest high-priority packet available, no Class 1 packets)
2	2	Class 2, arrival time 0.85 (the next Class 2 packet in arrival order)
3	6	Class 1, the only Class 1 packet (highest priority)
4	3	Class 2, arrival time 1.25 (the earliest remaining Class 2 packet)
5	5	Class 2, arrival time 1.97 (the next remaining Class 2 packet)
6	7	Class 2, arrival time 2.93 (the next remaining Class 2 packet)
7	10	Class 2, arrival time 3.95 (the last remaining Class 2 packet)
8	4	Class 3, arrival time 1.68 (the earliest Class 3 packet)

Time Slot t	Transmitted Packet	Reason
9	8	Class 3, arrival time 3.24 (the next Class 3 packet)
10	9	Class 3, arrival time 3.55 (the last Class 3 packet)

(2) Round-Robin Scheduling (Cycle: Class 1 → Class 2 → Class 3)

Scheduling Rule: Cycle through classes in the order of Class 1 → Class 2 → Class 3. Transmit the earliest arrived packet of the current class if available; skip the class if no packets are queued.

Time Slot t	Transmitted Packet	Reason
1	1	Cycle to Class 1 (no packets) → Class 2 (has packets, earliest is 1)
2	2	Cycle to Class 3 (no packets) → Class 1 (no packets) → Class 2 (next is 2)
3	4	Cycle to Class 3 (has packets, earliest is 4)
4	6	Cycle to Class 1 (has packets, only 6)
5	3	Cycle to Class 2 (earliest remaining is 3)
6	8	Cycle to Class 3 (earliest remaining is 8)
7	5	Cycle to Class 1 (no packets) → Class 2 (earliest remaining is 5)
8	9	Cycle to Class 3 (earliest remaining is 9)
9	7	Cycle to Class 1 (no packets) → Class 2 (earliest remaining is 7)
10	10	Cycle to Class 3 (no packets) → Class 1 (no packets) → Class 2 (last remaining is 10)